

Predicting High-Cost Medicare Patients

Before They Become High-Cost

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5%

of patients drive
50% of costs

72%

of future high-cost
patients captured

Your ACO Is Flying Blind on Next Year's Costliest Patients

5%

of Medicare patients
drive 50% of all costs

\$85,798

annual spending threshold
for top-5% patients in 2023

\$0

proactive identification
in most ACOs today

By the time a patient is identified as high-cost, the hospitalizations have already happened, and the savings window has closed. Your care management team needs a forward-looking tool that ranks patients by predicted risk before the year begins.

Our answer: Use last year's Medicare claims to predict who will be in the top 5% of spending next year, and flag them for proactive outreach before avoidable hospitalizations occur.

What We Set Out to Do — and For Whom

OBJECTIVE

Predict which Medicare patients will be in the top 5% of spending in Year 2 using only Year 1 claims data

TARGET POPULATION

Fee-for-service Medicare beneficiaries with 12 months of Part A & B enrollment. Medicare Advantage excluded.

DECISION / USE CASE

Ranked risk-score list delivered each January so care managers can prioritize outreach before costs materialize

INTENDED AUDIENCE

ACO Medical Director and care coordination team allocating limited outreach capacity across a large panel

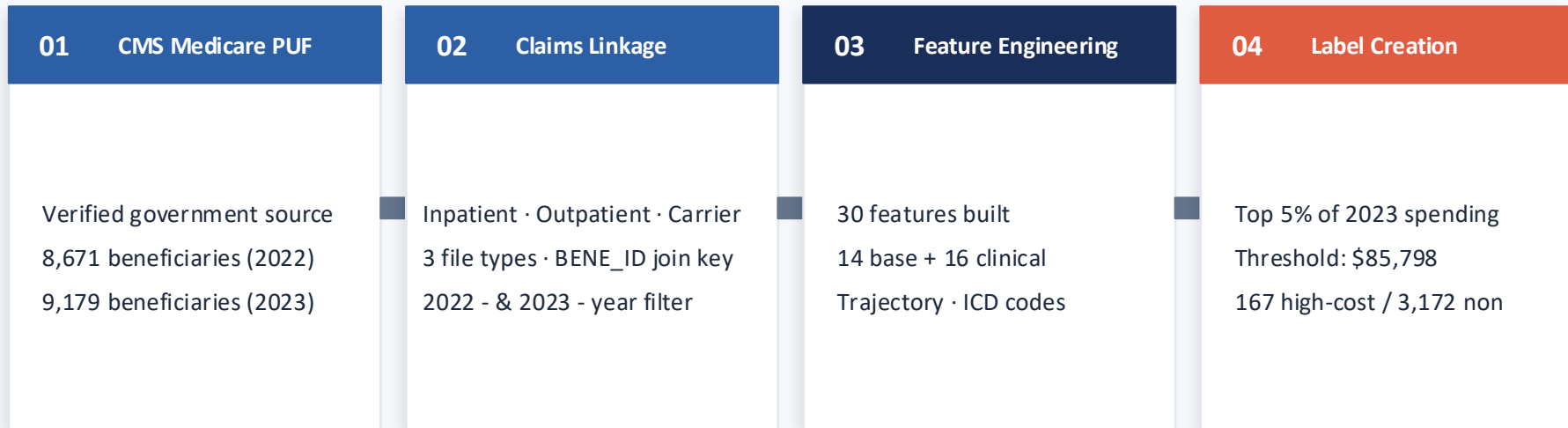
EVALUATION FOCUS

AUC-ROC: measures how well the model ranks patients by risk, regardless of class imbalance

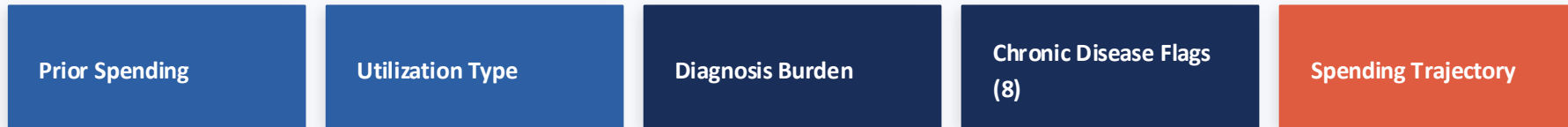
BOUNDARIES

Identification only. Tool informs clinical judgment, it does not replace it or automate any care decision

From Raw Claims to a Prediction-Ready Dataset



30 Features Across 5 Categories:



The Data Already Tells a Story

13x Spending Gap

Future high-cost patients had mean Year 1 spending of \$1,160K vs \$88K for non-high-cost patients a 13x differential that validates prior spending as the dominant predictor.

Right-Skewed Costs

Year 2 spending is heavily right-skewed. Most patients cluster near zero while a tiny tail drives disproportionate cost confirming the top-5% concentration pattern.

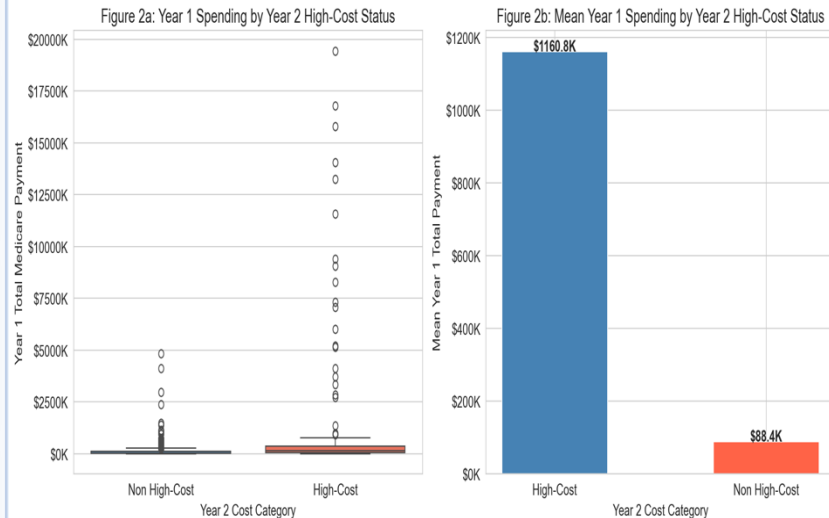
Demographics Signal

Males have a 5.8% high-cost rate vs 4.3% for females. High-cost patients peak younger (~65) than non-high-cost (~70), reflecting acute catastrophic events.

Disease Prevalence

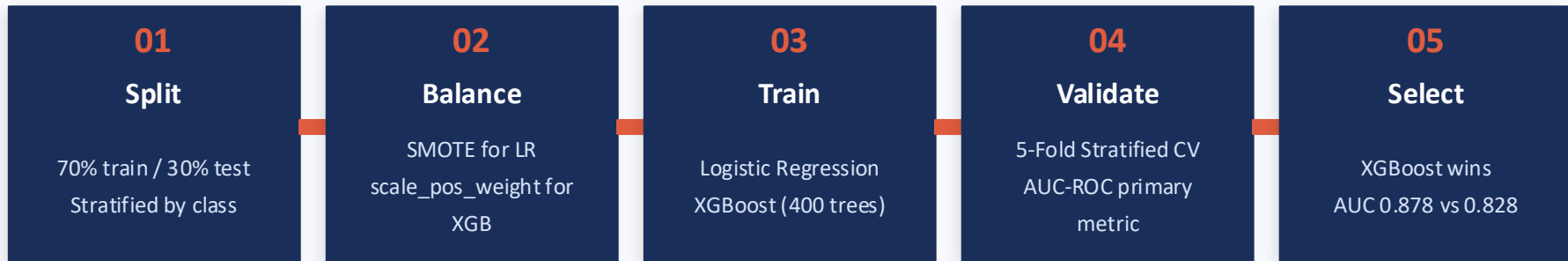
41.5% cardiovascular, 36.8% renal, 32.8% diabetes, clinically realistic Medicare population with high chronic disease burden.

Year 1 Spending Patterns by Future High-Cost Status



Prior-year spending is the clearest signal, and it means we can identify high-risk patients using data ACOs already have.

How We Built and Selected the Final Model



Why XGBoost Over Logistic Regression?

- Captures non-linear interactions between features (e.g. high spending AND high growth rate = far higher risk than either alone)
- Handles class imbalance natively via scale_pos_weight, no synthetic data needed
- Test AUC 0.878 vs 0.828 for Logistic Regression, clear winner on the primary metric

Model	Features	Test AUC
LR Baseline	14	0.828
XGB Baseline	14	0.796
LR Enriched	30	0.828
XGB Enriched ★	30	0.878

How Well Does the Model Perform?

AUC-ROC

0.878

vs 0.828 baseline
+5 point improvement

AVG PRECISION

0.592

vs 0.050 random
12× better than chance

CAPTURE RATE

72%

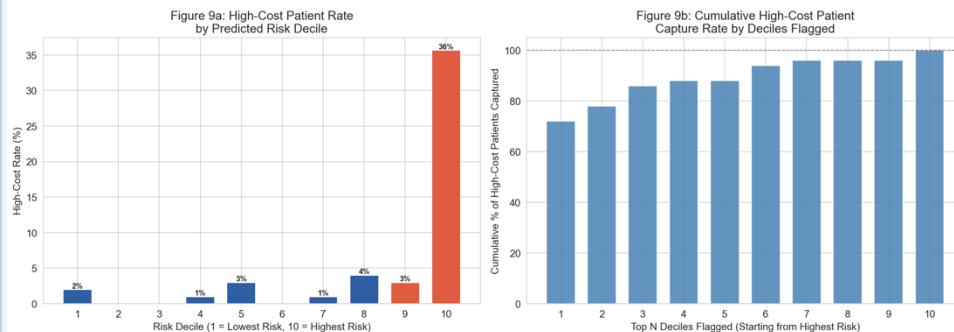
Flagging top 10%
of patients

SPENDING COVERED

56%

Top decile accounts
for 56% of all costs

Figure 9: Risk Decile Analysis — XGBoost Enriched Model

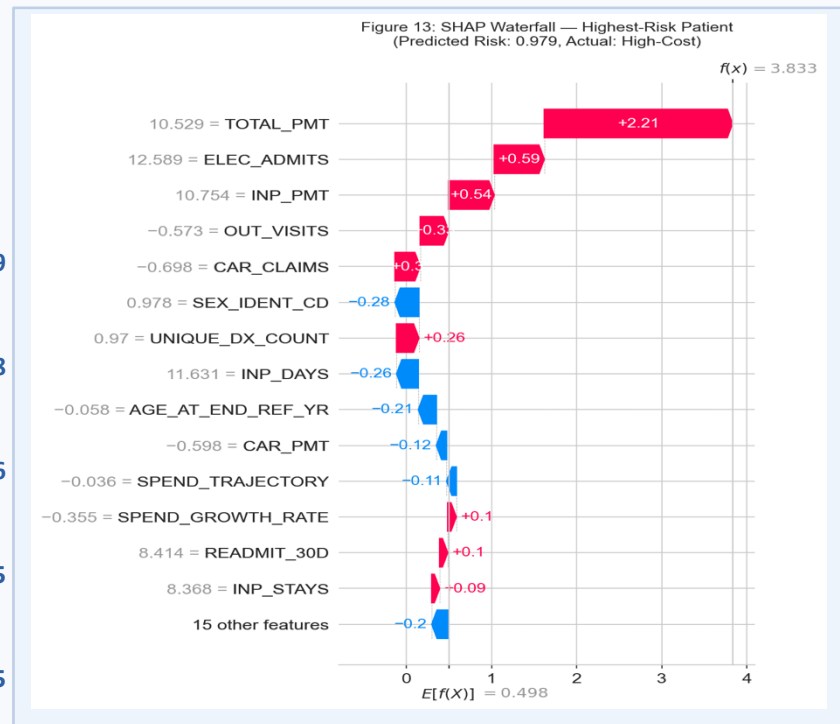
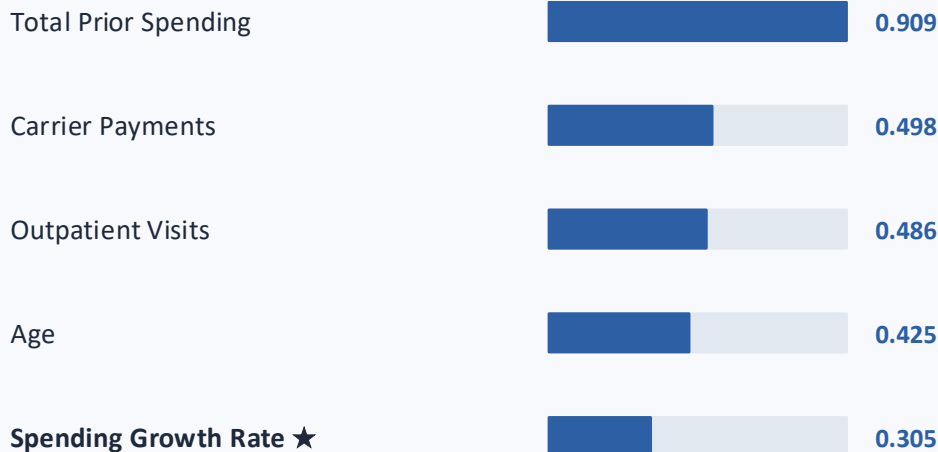


Our model beats published benchmarks (Chechulin 2014: AUC 0.78–0.82) and delivers operationally actionable results.

Why Is Each Patient Flagged?

SHAP explains each individual prediction — not just which features matter globally, but which factors drove THIS patient's risk score.

TOP 5 PREDICTORS (Mean |SHAP|)



When your nurse calls a flagged patient, SHAP tells them exactly why — enabling a targeted, personalized conversation.

Three Steps to Deploy This Today

01

Run the Model Each January

Apply the model to your full enrolled Medicare panel using the prior calendar year's claims. Generate a ranked risk-score list before the care management planning cycle begins.

Proactive — not reactive**02**

Flag the Top 10%, Not Just Top 5%

On a 5,000-member panel: 500 targeted contacts. Broader net buffers for model uncertainty while capturing 72% of future high-cost patients and 56% of all projected spending.

500 contacts → 72% captured**03**

Pair Every Flag With Its SHAP Explanation

A patient flagged for high inpatient spending needs a different conversation than one flagged for rising specialist visit frequency. SHAP makes every outreach call smarter.

Personalized intervention

Preventing even 15% of avoidable hospitalizations among flagged patients generates ROI that substantially exceeds program cost — and for your most vulnerable members, earlier intervention is not just financial, it is clinical.

THE BOTTOM LINE

Flag 10% of your panel. Capture 72% of next year's highest-cost patients.

Using only claims data your ACO already has.

0.878

AUC-ROC

72%

Capture Rate

56%

Spending Covered

Limitations & Next Steps

- Sample: 3,339 patients, validate on full panel before deployment
- Retrain model annually as claims patterns evolve
- Add calibration correction (Platt scaling) for probability reliability
- Extend to chronic condition sub-groups for precision targeting